

Building & Thermal Inspection Report

Property: 123 Innovation Drive, San Francisco, CA

Date: May 29, 2026

Inspector: J. Smith

Client: Acme Property Management

1. Executive Summary

This report details findings from a comprehensive property and thermal inspection. Several moderate-to-high severity issues were identified in the electrical panel and HVAC ductwork. Immediate attention is recommended for two thermal anomalies detected in the main distribution board.

2. General Observations

Structural: The building frame is in good overall condition. Minor cracking observed in the southeast corner drywall (non-structural).

Roofing: Flat membrane roof shows pooling near drainage outlet B. Membrane integrity is satisfactory but drainage slope requires correction.

Electrical: Main distribution panel shows signs of prior water ingress. Breaker labeling is incomplete. Two breakers show abnormal thermal signatures.

HVAC: Rooftop unit RTU-2 shows refrigerant line insulation degradation. Supply duct in the basement shows a 4°C temperature drop indicating air leakage.

3. Thermal Findings

Finding T-01 — High Severity

Location: Main Electrical Panel — Breaker L3

Temperature: 68.2°C (ambient 24°C)

Description: Significant hotspot detected on breaker L3. Delta-T of 44°C indicates loose connection or overload. Immediate investigation required to prevent fire risk.

Finding T-02 — Medium Severity

Location: HVAC Supply Duct — Basement Run

Temperature: 12.8°C (supply air setpoint 16°C)

Description: Cold spot along duct seam indicates air leakage. Energy loss estimated at 15%. Recommend duct sealing.

Finding T-03 — Low Severity

Location: Exterior Wall — North Elevation

Temperature: 18.5°C (interior 22°C)

Description: Minor thermal bridging at window frame. Insulation gap detected. Low priority repair.

4. Risk Assessment

Overall risk score: **6.8 / 10 (Medium-High)**

Safety risk is elevated due to electrical hotspot (T-01). Structural risk is low. Operational risk is moderate due to HVAC inefficiency.

5. Recommendations

High Priority:

- Hire licensed electrician to inspect and replace breaker L3 and tighten bus connections.
- Install continuous thermal monitoring on main panel until repairs are completed.

Medium Priority:

- Seal basement supply duct seams with mastic and wrap with new insulation.
- Correct roof drainage slope and clear outlet B debris.

Low Priority:

- Repatch drywall cracks in southeast corner.
- Replace missing outlet cover plates on Level 2.

6. Missing Information

- As-built electrical drawings not available.
- Previous HVAC maintenance records were not provided.
- Roof warranty documentation missing.